A Claim-Admission Protocol for Diffusion Membership-Inference Scores: Boundary Cases and Evidence States

Anonymous Authors

***Abstract*—Diffusion membership-inference attacks are usually reported as scalar scores, but a score alone does not determine what claim it can support. The same high-AUC signal may be genuine membership evidence, a covariate artifact, a non-portable response-surface effect, or a finite-tail measurement too weak for forensic use. We propose a claim-admission protocol for diffusion MIA scores: each reported claim is bound to a target identity, split semantics, row-level observable, metric provenance, finite-tail basis, consumer boundary, and surface-delta requirement, and is assigned the narrowest evidence state supported by those surfaces.**

Across 24 boundary-case rows, the protocol separates admitted, candidate, support-only, spurious, weak, and non-portable evidence. The central case is an activation-subspace fingerprint scout, H1, which is not a trivial failure. Internal U-Net activations carry replicated aggregate membership signal across two same-architecture checkpoints, with AUC 0.873 on DDPM-800k and 0.841 on DDIM-750k, and shuffle controls near chance. The signal is dominated by activation magnitude. Yet channel knockout experiments show it is distributed and non-localizable: removing the most membership-correlated channels does not suppress the score more than matched random deletion (Cohen's d=0.10–0.21, p=0.26–0.60 across checkpoints). An 8-timestep fine temporal grid further reveals that membership signal geometry is training-procedure-dependent: DDPM stochastic training produces temporally distributed signal (max individual $\triangle$AUC=0.029), while DDIM deterministic training produces temporally concentrated signal ($\triangle$AUC=0.221, a 7.6$\times$ difference)—the only DAAB property that varies across checkpoints. At the same time, H1 is not fully forensically admitted: on DDPM, TPR@1%FPR collapses from 0.484 (N=64) to 0.055 (N=128); on DDIM, TPR@1%FPR maintains 0.273 at N=128 but TPR@0.1%FPR remains 0.000 for both checkpoints. The forensic tail is thus partially training-dependent: DDIM preserves moderate low-FPR recovery where DDPM loses it, but neither checkpoint supports sub-percent FPR claims.

We term this pattern Distributed Activation-Amplitude Bias (DAAB): a real, replicated, mechanistically characterized membership trace that is channel-level distributed and universal across training procedures, temporally distributed or concentrated depending on the training objective, and too fragile at low FPR for forensic admission. Additional boundary cases include CLiD (spurious: AUC 1.000 collapses to 0.586 under prompt-neutral control; $\triangle$AUC reported as point estimate only), scnet (scale-null: 54$\times$ capacity, $\triangle$AUC=0.003), H2 output-cloud (non-portable: AUC 0.962 within-family, fails img2img), and MoFIT (external support: AUC 0.942, missing row binding). The broader lesson is that membership inference is not merely a score-ranking problem: a diffusion MIA score becomes evidence only after its claim boundary is admitted.

I. Introduction

Diffusion models can memorize or replicate training examples in ways that enable membership inference attacks [1], [2]. The literature has produced black-box, gray-box, and white-box signals, including reconstruction similarity, posterior-error statistics, proximal initialization, conditional likelihood discrepancy, black-box reconstruction, and white-box gradient attacks [3], [4], [5], [6], [7]. We ask what audit claim a diffusion MIA score packet can support. The same score can serve as a replay anchor, a candidate signal, or a blocked claim depending on which evidence surfaces are present. A model owner, replay auditor, or downstream report consumer needs to know which target checkpoint was audited, what row IDs define members and nonmembers, whether scores exist per sample, how low-FPR tails should be interpreted, and which claims are blocked by missing artifacts.

We use a claim-bound evidence packet to carry those fields. The audit object is a structured packet with defined target, split, observable, metric, consumer, and delta surfaces. Following security-measurement and reproducibility norms [8], [9], [10], the contract specifies target identity, split semantics, score or response coverage, metric provenance, consumer boundary, and surface delta as gates. Given frozen gate labels, a rule maps each claim to an audit state. The claim register exposes the gate labels, the first missing surface, and the allowed wording for each checked claim. The measured object is claim-support eligibility; attack-method ranking is a separate benchmark problem. The resulting table states the audit wording each concrete packet can support under a fixed evidence contract.

This paper defines the packet fields required for model-owner, replay-auditor, and downstream-report claims. It applies those fields to score/response packets and to high-score or metadata-only boundary rows, recording which gate blocks which claim wording. The contribution is not a new attack or defense, but a measurement framework that separates what a score *is* (a scalar) from what a score *supports* (an evidence state). The H1/DAAB case demonstrates this separation empirically: real, replicated, mechanistically characterized signal that is non-localizable at the channel level and forensically fragile.

The contributions are:

- A claim-admission protocol with six gated surfaces: target identity, split semantics, score/response coverage, met-

ric provenance, consumer boundary, and surface delta. The transition rule maps a proposed audit wording to reportable, candidate, support-only, watch, or negative states by recording the first missing surface and the narrower wording the packet supports.
- A conceptual admission map (Fig. 1) showing that high aggregate signal strength does not imply claim-boundary validity: real, replicated signals can remain non-localizable and forensically fragile.
- We apply the protocol to replay positives, point comparators, metric-only negatives, and source/artifact stops. The resulting trace records the missing surfaces behind interval, dominance, and leaderboard claims.
- H2 output-cloud reaches AUC 0.9615; unresolved consumer-boundary and surface-delta gates keep it candidate-only. Same-family metric checks leave the row in candidate state.
- Under the protocol labels, a public score-file boundary case (MoFit, AUC 0.941948) is support-only. Public score-like files lack immutable target identity, explicit row IDs, certified row binding, consumer-boundary support, and surface delta.
- A C14 stress packet shows how weak public-surface rules over-select pre-label rows: code and paper-artifact links appear in 12/13 selected rows, while all selected rows are non-Pass at the score/response and consumer-boundary gates. The packet supports pre-label stress-control wording. The separate 28-case release-wording matrix handles phrase-guard regression wording.

II. Related Work

A. Membership Inference and Evidence

Membership inference asks whether a record was used during training [11]. Prior work links this risk to overfitting and to the statistical meaning of membership scores [12], [13]. Recent position work warns against treating MIA as standalone proof of an individual datum's use [14].

Training-data-use claims need evidence beyond MIA scores. The protocol scopes score-backed audit wording around that gap. Foundation-model audits also show that blind baselines can outperform MIA attacks when member and nonmember sources differ for reasons unrelated to membership [15]. A claim-admission protocol then fixes the required surfaces: target, split, row-level observable, metric provenance, and consumer boundary must be explicit before wording can strengthen.

B. Security Measurement and Audit Claims

Security/privacy measurement differs from attack benchmarking because the measured unit, denominator, replay surface, and downstream consumer determine what a number can mean. Security-science and benchmarking work similarly argues that scientific security claims require explicit evidence units, denominators, measurement assumptions, and reproducible comparisons [8], [9]. For MIA, first-principles metric analysis shows that accuracy and tail rates require careful base-rate and threshold interpretation [13]. Recent audit-oriented interpretations separate membership evidence from proof of training-data use [14].

For diffusion MIA, the claim-admission protocol records which target/split/row observable and replay tier make a score consumable. Partial packets stay as candidate or support evidence outside a leaderboard. Attack papers provide evidence surfaces and packets; the protocol assigns claim-support states under a fixed contract.

C. Diffusion Memorization and MIA

Diffusion models [16], [17] expose privacy risk through memorized generations and membership signals [1], [2]. The access surface matters: gray-box posterior, trajectory, and likelihood surrogates [3], [4], [5], [18], black-box response or reconstruction methods [19], [6], caption-free fitted-embedding scores [20], and white-box gradient attacks [7] support different audit claims. Recent initial-noise probing expands this taxonomy to noise-schedule surfaces, reinforcing that modality-specific signals should be gated separately [21]. DPDM, CopyMark, MIDST, and CDI further show that defense, benchmark, and dataset-level contracts change what a result can mean [22], [23], [24], [25].

CopyMark [23] is the closest prior work in spirit: it shows that published diffusion MIA numbers often degrade under realistic benchmarks with pre-trained models, unbiased datasets, and fair pipelines, and it emphasizes low-FPR evaluation. Our work addresses a complementary question. CopyMark asks whether MIAs survive more realistic benchmarks; we ask what conditions must be satisfied before any MIA score—even a reproducible, statistically non-random one—is admissible as evidence for a specific claim. CopyMark's unit of analysis is the attack method on a benchmark; ours is the score surface under an evidence contract with named claim boundaries. The two contributions are compatible: CopyMark supplies realistic evaluation conditions, while DiffAudit supplies the admission gates that decide what a score measured under those conditions can support.

D. Reporting and Reproducibility Artifacts

Reproducibility checklists, datasheets, and model cards document data, code, conditions, origins, limitations, and intended use [10], [26], [27]. We evaluate claim admissibility at the row-bound packet level. Row-level membership labels or responses, metric provenance, finite-tail denominators, replay tier, surface delta, and allowed consumer language decide what a number can mean. Claim state is the narrowest wording allowed by row-level provenance, finite-tail basis, consumer action path, and code availability as separate surfaces.

The decision trace records an ordered gate vector, the first non-Pass blocker, and the narrower wording allowed after that blocker is named.

III. Audit Surfaces

Audit admission asks which claims a score packet can support. The membership game is binary: for a fixed target and

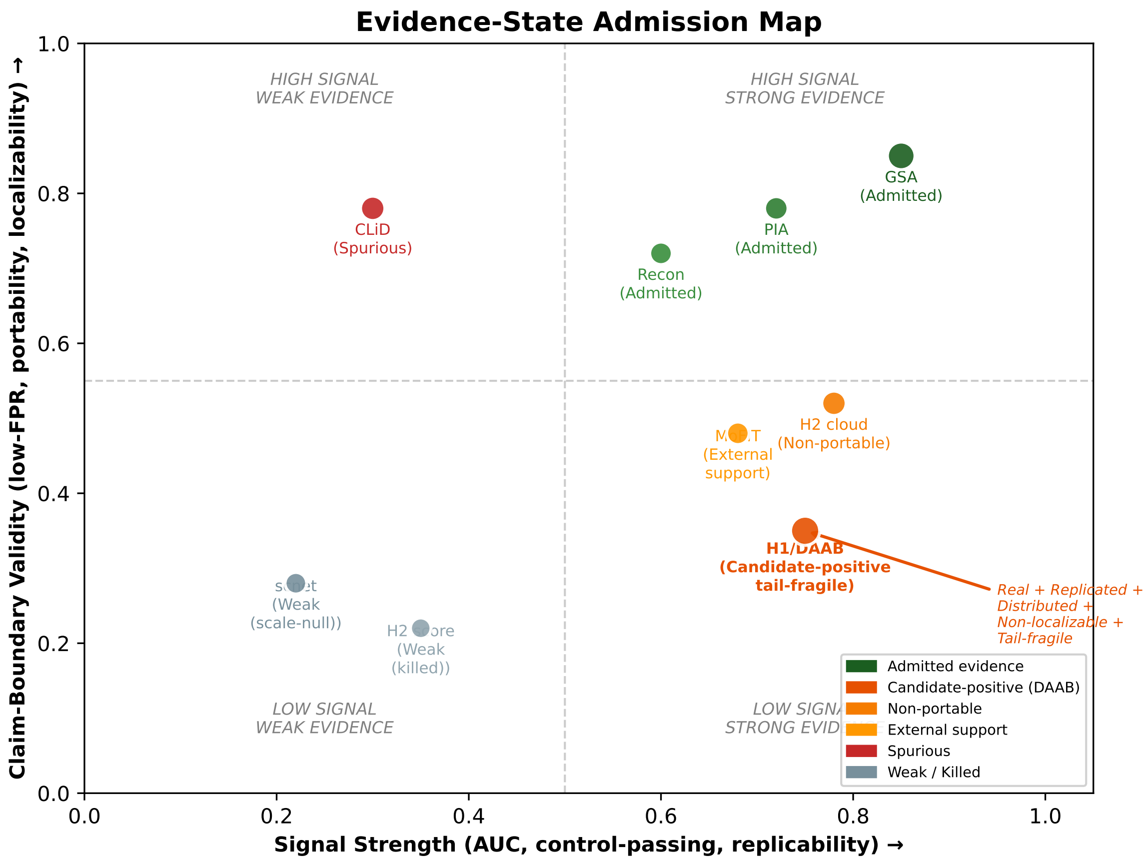

Fig. 1. **Score strength is not evidence admission.** High AUC moves a score rightward, but only provenance, calibration, controls, and portability move it upward into admissible evidence. GSA achieves both high separability and admissible claim status. H1/DAAB achieves high separability but remains tail-fragile and non-localizable, occupying the lower-right quadrant. CLiD and H2 achieve high AUC under narrow conditions but fail under controls or portability tests. The evidence-contracted audit maps each score packet to its admissible claim region.

frozen member/nonmember split, an evidence packet scores the fixed benchmark member/nonmember label under a stated target, split, and access mode. The audit output is the claim this packet can support.

The protocol reports packet-level claim support. Leaderboard normalization, individual training-data proof, field-wide artifact prevalence, and consumer-facing candidate-row claims require separate measurements. The split gate requires fixed member and nonmember definitions.

We group prior diffusion MIA artifacts by access surface [3], [4], [5], [19], [6], [7]. Each access surface supports a distinct claim role. A black-box response packet can support an output-consumer claim when target identity and query rows are fixed. A gray-box trajectory or posterior-statistics packet can support a mechanism or evaluator claim. A white-box gradient packet is an upper-bound risk comparator. Feature packets and separately supplied local scores support mechanism analysis or hypothesis generation and remain support-only as audit evidence. Table I records the surfaces used in this paper. Each access mode binds its own evidence surface. A row reported under one surface requires independent gate evaluation under another.

TABLE I
AUDIT SURFACES AND ALLOWED EVIDENCE ROLES.

Surface	Allowed paper role
Black-box responses	Reportable only with fixed target, split, response coverage, and metric replay.
Gray-box trajectory/statistics	Reportable or candidate depending on provenance and consumer semantics.
White-box gradients	Upper-bound comparator unless the audit consumer has white-box access.
Feature packets	Support-only unless raw target/sample provenance and regeneration assets are available.
Restricted or separately supplied packets	Stress tests unless public row binding and source-confounding checks pass.

IV. EVIDENCE CONTRACT

The protocol assigns claim state with six gates: target identity, split semantics, score/response coverage, metric provenance, consumer boundary, and surface delta. Target identity fixes the model architecture, checkpoint identity, training provenance, and accessibility. Split semantics fixes member and nonmember manifests. Score/response coverage binds row-level outputs to labels. Metric provenance records AUC, ASR, TPR@1%FPR, TPR@0.1%FPR, nonmember denominator, and replay tier: row-score replay, response/feature replay, target-score replay, or source-documented point estimate. Consumer-boundary checks expose the artifacts needed by the named consumer before wording is assigned. Surface delta checks whether a claimed second asset changes model, data, modality, or response contract. Each gate corresponds to a failure mode observed in the boundary cases: unbound checkpoint claims, source-pool shortcut membership, code-only or copied-metric promotion, high-score claims without an action-compatible consumer surface, and adjacent controls rewritten as portability evidence.

Consumer requirements are fixed before gate labels are assigned. We use three consumer templates: model-owner risk reporting, replay audit, and downstream summary. Table II records the artifacts each task must expose. A consumer-boundary failure means the packet lacks the artifacts for the named task.

Under this contract, code, checkpoints, score files, and high candidate scores support downstream audit wording after the packet exposes target, split, metric, consumer, and surface-delta surfaces. The contract governs audit wording, not full scientific value or reproduction scope.

Low-FPR metrics are reported as finite empirical packet readouts [13]. For example, TPR@0.1%FPR on 100 nonmembers has a false-positive budget of 0.1; zero observed false positives are the only admissible count under the strict empirical rule. Small-packet tail values are finite empirical readouts; calibrated sub-percent operating-point language overstates them.

Gate labels are claim-relative measurement decisions. *Pass* means the packet exposes the surface required for the stated claim. Replay claims require row-bound score, response, or target-score arrays. Point claims require a frozen source-documented metric and narrower wording. *Partial* means a useful surface exists and a stronger claim still needs more identity, provenance, or consumer binding; *Fail* means a missing or contradictory surface blocks that claim; and *N/A* means the gate is outside the claim type. Each label records the missing surface that would change the state, so the protocol records an audit decision. A metric gate pass carries a replay tier. Point-estimate passes support exact bounded point wording; interval and dominance claims require direct score arrays.

Consumer-boundary labels identify claims that depend on hidden data, new execution, or a different evidence surface.

V. Measurement Protocol

Under fixed protocol labels, each result receives one of five evidence states. A reportable row can be used for contracted reporting under its stated boundary and caveat. A candidate row supports follow-up analysis. A support-only row motivates or contextualizes a mechanism. A watch item is tracked for future artifacts. A negative row answers a bounded decision question and usually closes a low-value expansion.

We formalize the protocol as a finite-state gate-decision rule over a proposed audit claim and its supporting packet. Its input is an audit claim template c and an evidence packet $p = (T, S, O, M, C, \Delta)$ binding target T, split S, row-level observable O, metric provenance M, consumer boundary C, and optional surface delta Δ. It evaluates the ordered gate vector $g(p, c) = \langle g_T, g_S, g_O, g_M, g_C, g_\Delta \rangle$ and transitions to the narrowest state justified by the first required non-Pass gate. Reportability requires target, split, evidence, metric, and consumer gates to pass for the proposed wording; the delta gate must pass or be irrelevant to the claim. Replay admission requires a metric tier with row, target-row, response, or feature replay. A frozen source-documented point estimate can be reportable for exact bounded point wording. Figure 2 makes the transition rule explicit.

TABLE II

Evidence-contract gates as protocol checks. Consumer-boundary rows make the downstream action and missing artifact explicit before wording is assigned.

Gate	Required surface checked by the protocol
Target identity	Fixed architecture, checkpoint, training provenance, or target descriptor.
Split semantics	Immutable member/nonmember row manifests or split definition.
Score/response coverage	Row-level scores, responses, features, or metric packet for labeled samples.
Metric provenance	Replay tier: row-score, response/feature, target-score, or documented point estimate.
Consumer boundary: model owner	Can name target/checkpoint, split, access mode, operating point, finite-tail denominator, and owner action for the row role.
Consumer boundary: replay auditor	Can recompute or verify from row-score, target-score, response, or feature packet plus metric code/JSON; source points stay point-only.
Consumer boundary: downstream report	Can quote target/split, replay tier, provenance, denominator, and caveat in a bounded sentence; hidden fields block audit wording.
Surface delta	Evidence that a claimed second asset changes model, data, modality, or response contract and is distinct from the adjacent surface.

The coded claim register is the claim-state table: each checked claim maps to one first blocker, state, replay tier, and allowed sentence. It assigns dominance, interval, portability, prevalence, and route-generalization wording from the recorded state and provenance. This register check covers selected boundary cases. General gate sufficiency and independent-coding reliability require broader corpora and independent labels.

The supplementary material includes a four-row transition slice derived from the trace. It exposes reportable, candidate, uncertainty-sidecar, and C14 packet-ready cases without requiring the full register.

Metrics are computed from row-level scores or frozen response packets; otherwise they are retained as declared point estimates. We report AUC, attack success rate (ASR), TPR@1%FPR, and TPR@0.1%FPR. Low-FPR values are interpreted against the nonmember denominator. The implementation takes the maximum empirical TPR at thresholds with FPR at most the target; when $0.001 n_0 < 1$, TPR@0.1%FPR is a zero-FP tail on a discrete grid. The trace records both metric derivations and packet boundaries.

Gate-assignment procedure.

1) Bind T, S, O, M, C, Δ from public packet surfaces and source provenance before reading aggregate counts.
2) Evaluate gates in fixed order: target, split, evidence, metric, consumer boundary, and delta. Assign Pass only when the surface needed by claim c is visible in the packet; assign Partial when a related surface is visible and the proposed wording needs another surface; assign Fail for missing or contradictory required surfaces; assign N/A only when the claim type does not use that gate.
3) Record the first non-Pass required gate as the first blocker. Later passes do not override this blocker.
4) Emit the narrowest allowed state: reportable for all required Pass gates, candidate for unresolved consumer or portability gates, support-only for different access or feature surfaces, watch for artifact discovery without row binding, and negative for a bounded route-selection stop.
5) Attach replay tier. Row-score, target-score, response, and feature replay tiers can be replay-admitted; source-documented point estimates remain point-worded.

Fig. 2. Fixed-order gate-assignment procedure used by the coded claim register.

A. Metric Replay

Metric replay is a provenance check. When a score array exists, metrics are recomputed from row labels. When only responses exist, the scorer and response manifest must be fixed before the metric can be used. For a nonmember packet of size n_0, FPR takes values on a discrete grid of $1/n_0$. TPR@0.1%FPR is a strict empirical tail whenever $0.001n_0 < 1$. We report the readout as a finite packet comparison and retain the denominator; calibrated-risk language would overstate the measurement resolution [13].

Table III is used later as a claim-admission diagnostic. Code or metadata availability leaves row binding, metric replay, consumer boundary, and surface delta gates unresolved. Rows that fail these gates remain blocked regardless of code or metadata presence.

The matrix identifies which missing surface blocks score-only, code-only, and replay-only claims. The contract ties each allowed sentence to the first blocker, replay tier, claim trace, source provenance, and metric-uncertainty sidecars.

VI. Boundary-Case Trace and Claim Outcomes

For selected boundary cases, the coded gates record the evidence state plus diagnostic tag: reportable, candidate, support-only, watch, or negative, with known-surface miss, source-confounded, or metadata-only where applicable. The trace also marks rows where weak admission rules would admit a claim that the contract blocks. Table IV fixes the selected row sets before counts are read. The declared set is a fixed boundary-case set. It tests claim-state behavior across bounded positives, a high-score metric-only non-admission case, route closures, and L0/L1 artifact stops. For frozen gate labels, revision is triggered when a blocked row has all five required surfaces: fixed target identity, fixed membership semantics, row-bound scores or responses, replayable metrics, and a compatible consumer boundary. Revision is also triggered when the first blocker is not reproducible from the packet.

Rows enter when they instantiate at least one gate-relevant surface or blocker before aggregate counts are read. The set has five control roles. Reportable rows are replay/source-tier anchors. H2 is a metric-strength boundary case: high AUC with failed consumer and delta gates. ReDiffuse STL-10, CommonCanvas, and MIDST are route-closure checks. Tracing the Roots and the separately supplied Stable Diffusion packet test support-only and source-confounded states. Fixed-search, source-query, and targeted-link rows test row-set discipline and L0/L1 artifact stops.

Decision-affecting artifacts entered through an evidence source, released trace entry, or machine-readable artifact before count readout. Duplicates are recorded as deduplication or screening rows. Background literature, artifact-discovery notes, large new-execution assets, and citation-only rows are outside the declared set. Later label changes are reported as label-review resolutions or source-query control updates. The selected rows show how the protocol accepts, narrows, or blocks fixed claim wordings. Their role is boundary checking; frequency estimation requires separate corpus sampling.

Each C1–C15 claim state has an inspectable first missing surface in the coded claim register. For every C1–C15 claim, the trace and provenance sidecars list the gate labels, first blocker, replay tier, state, allowed wording, source hashes, commit identity, generator command, and availability tier. Asset generation checks that provenance IDs resolve and gate labels use allowed values. For the current packet, these fields identify the local review snapshot; the public-release provenance gate remains unresolved.

A. Boundary-Case Trace

The declared set was built from audited evidence surfaces and frozen before count readout. Section XI separates these boundary-case labels from corpus-scale and coding-reliability claims.

The supplement provides a shuffled blank C1–C15 claim-gate recode template, a hash manifest, and a 28-case release-wording regression matrix. The template carries blank author-gate fields. Independent recode labels and reliability results have not been collected. The regression matrix tests the admitted-only risk-card renderer and candidate-promotion phrase guard as release QA for wording boundaries. Report drift, compute-release gate validity, and external-use estimates remain outside this packet.

The 2026-05-26 metadata search used seven GitHub queries and five arXiv API queries, with no downloads or GPU runs. It records three re-found surfaces, one new code-public

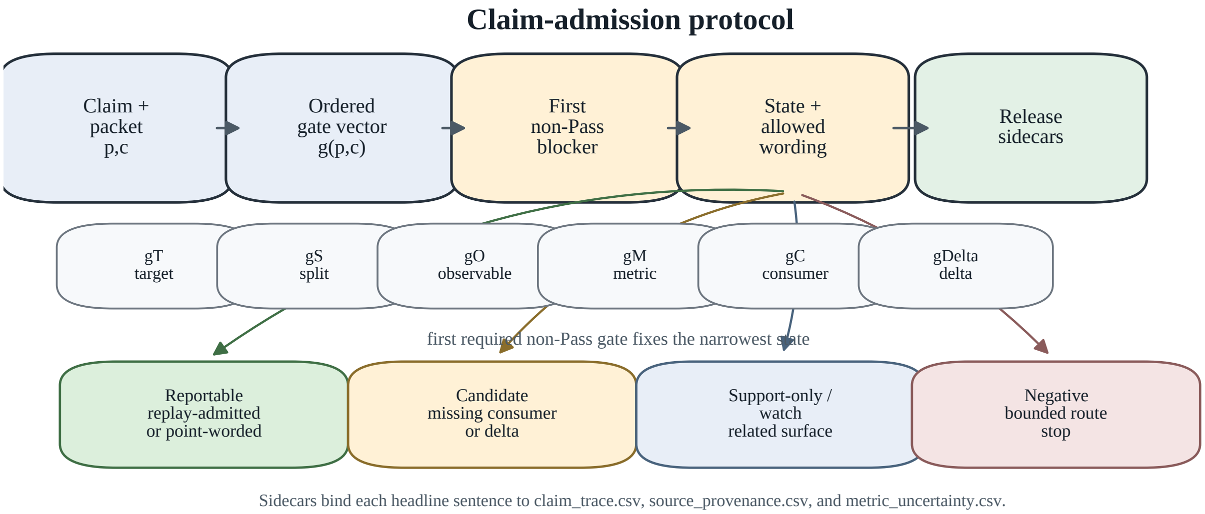

Fig. 3. Claim-admission protocol. The ordered gate vector and first required non-Pass blocker route a proposed claim to reportable, candidate, support-only/watch, or negative wording; release sidecars bind the resulting sentence to trace, provenance, and metric files.

TABLE III
CLAIM-ADMISSION GATE MATRIX EXAMPLES. LABELS ARE CLAIM-RELATIVE: A ROLE-PASS IS SUFFICIENT FOR THE STATED PAPER ROLE; DOWNSTREAM ADMISSION REQUIRES THE MATCHING CONSUMER AND SURFACE-DELTA FIELDS. REPORTABLE ROWS ARE SPLIT BY REPLAY TIER TO SEPARATE SOURCE-DOCUMENTED POINT ESTIMATES FROM ROW-SCORE REPLAY.

Route	Target	Split	Evidence	Metric	Boundary	Delta	Replay tier	Paper use
PIA rows	Pass	Pass	Pass	Pass	Pass	N/A	row-score	Row-replay gray-box audit and defense-comparator rows
Recon/GSA rows	Pass	Pass	Pass	Pass	Pass	N/A	src point	Black-box proxy and white-box upper-bound comparator rows
DPDM W-1 bridge	Pass	Pass	Pass	Pass	Pass	N/A	target-score	Target-score defense bridge
H2 output-cloud	Pass	Pass	Pass	Pass	Fail	Fail	response	Candidate only; consumer and portability blocked
SD/CelebA img2img portability	Pass	Pass	Pass	Pass	Fail	Pass	response	Candidate boundary; portability admission unresolved
MoFit public score files	Partial	Partial	Partial	Partial	Fail	Fail	public scores	Support-only score-file boundary; row binding and consumer blocked
ReDiffuse STL-10 strict/overfit	Partial	Pass	Partial	Partial	Fail	Fail	local route	Strict route negative; overfit support-only
CommonCanvas response packet	Partial	Pass	Pass	Pass	Fail	Pass	response	Bounded negative
Tracing the Roots feature packet	Partial	Pass	Pass	Pass	Partial	Pass	feature	Support-only
Supplied SD ReDiffuse	Partial	Partial	Pass	Pass	Fail	Pass	supplied scores	Source-confounded support

no-packet surface, seven relevant metadata-only rows, one existing-watch metadata row, four empty fixed-query rows, and one noisy query row; all observations preserved admission decisions. The L0–L3 counts document surface hits for trace inspection: 21 selected-set rows at 8/3/9/1 and 17 fixed-search rows at 13/4/0/0. A targeted artifact-link sidecar adds three L1 public artifact surfaces. Row scores, ROC curves, metric JSON, and verifier packets are absent.

The frozen-gate cross-reference and second pass are release-consistency checks. They preserved admission states and tightened two labels (withdrawn DLM delta, Noise Aggregation target). The packet is frozen-gate release material. Standalone artifact claims depend on the corpus-scale and reviewer-label criteria in Section XI.

B. Author-Keyed False-Promotion Stress Controls

C14 is a thirteen-row author-keyed stress packet that tests whether weak public-surface rules over-promote rows before independent reviewer labeling. The packet currently has n_reviewers=0 and serves as pre-label stress control, not as a prevalence or reliability result. Across 13 selected no-download public-surface rows, code and paper-artifact links trigger weak-admission rules in 12/13 rows, but every row is non-Pass at the score/response and consumer-boundary gates. The full C14 table, figure, and row-level blocker breakdown are provided in the supplement.

The author-keyed blockers distinguish diffusion MIA evidence from adjacent tasks such as attribution, classifier defense, copying, prompt memorization, and dataset-level iden-

TABLE IV
FROZEN BOUNDARY-CASE LAYERS USED FOR CLAIM-STATE CHECKS.

Layer	Fixed row set and check role
Boundary-case records	21 rows: 11 previously reviewed evidence-surface rows plus 10 metadata-only expansions; positive, negative, support-only, and source-confounded controls; no new downloads or model runs.
Fixed metadata batch	2026-05-26 pass with 7 GitHub repository queries and 5 arXiv API metadata queries; 17 rows test metadata-only L0/L1 stops and refound/noisy surfaces; no row changed an admission decision.
Source-query controls	2026-05-27 metadata-screening HF/Zenodo/OpenReview API pass; 10 observations test query-result hygiene and non-poolable row-set discipline; no observation changed an admission decision.
Targeted artifact links	2026-05-27 chase over existing metadata-only seeds; 3 L1 public artifact surfaces, including 2 split surfaces; artifact-inspection stops only, with no score/metric replay or row-set pooling.
Claim scope	Counts support boundary inspection for this declared set only; field-prevalence, artifact-quality, and reproduction-rate estimates require separate sampling.

tification. Author-keyed target and metric blockers support packet-readiness and weak-rule stress-control wording. Reviewer agreement and prevalence require independent labels and a larger sampled corpus.

VII. Worked Examples of Protocol Application

Rows enter the paper through a fixed worked-example path. A row must bind labeled samples to a stated target, expose metric provenance, attach finite-tail denominators, and carry consumer-boundary language. The bundle has two evidence strengths: PIA, PIA-dropout, and DPDM W-1 are replay-admitted from row or target-score arrays in the audited artifacts. Recon and the main GSA row are source-documented point estimates. They support exact bounded role claims; interval, dominance, and leaderboard claims require row replay. Metric magnitude is reported as packet evidence. Admission follows reusable audit semantics. The five cases are target/split specific: recon DDIM uses CIFAR-10 public-100 point evidence; PIA and PIA+stochastic dropout use CIFAR-10 512/512 row replay; GSA uses CIFAR-10 1k/1k point evidence; and DPDM W-1 is target-score bridge replay.

Table V summarizes five worked-example cases across access modes. Black-box reconstruction and gray-box PIA baseline provide primary leakage-risk rows under their stated modes. Stochastic dropout and DPDM W-1 are reportable under defense-comparator or bridge wording. GSA remains a source-documented upper-bound point comparator.

The table is organized by claim role and replay tier. The rows are a set of role-separated worked examples. The table's provenance shorthand is part of the claim: *row* denotes row-score replay, *point* denotes source-documented point estimates without replay-derived intervals, and *target row* denotes target-score replay for the defended bridge. Gate labels are frozen protocol labels over selected rows.

AUC uncertainty is reported only where the replay surface supports it. The companion metric table gives bootstrap intervals for PIA baseline [0.8168, 0.8640], PIA-dropout

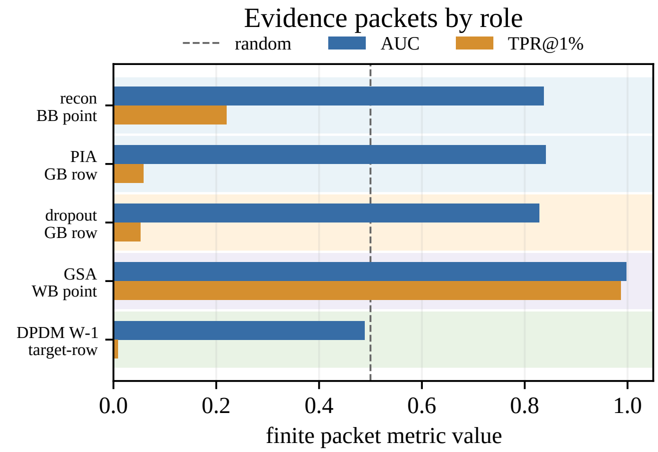

Fig. 4. AUC and TPR@1%FPR for role-separated worked-example cases. Rows are ordered by access and allowed claim role, and each label includes its replay tier. The figure catalogues role-separated evidence; benchmark ranking and replay-derived intervals are outside this display.

[0.8039, 0.8535], and DPDM W-1 [0.4640, 0.5152]. These intervals are AUC intervals under exchangeable-row bootstraps. TPR@1%FPR and TPR@0.1%FPR are denominator-and-count packet summaries. Recon and the main GSA row are point estimates, and H2 intervals are candidate-side controls. Each row supports the consumer claim named by its boundary or caveat.

VIII. Metric-Strength Boundary Case: H2 Output-Cloud

H2 is a high-AUC non-admission stress test. It scores output-output geometry from repeated generations and excludes query-to-output distance. Relative to black-box variation and reconstruction attacks that compare a query with returned or averaged outputs [19], [6], H2 isolates a response-contract observable. The packet stays candidate-only: consumer-boundary and surface-delta gates are unresolved. The fixed scorer featurizes within-query response clouds, fits a held-out logistic readout, and retains aggregate prediction summaries, coefficients, and recorded intervals; no new responses are generated for this paper.

On the existing 512/512 cache, output-cloud logistic scoring reaches AUC 0.9615, ASR 0.9004, TPR@1%FPR 0.3340, and TPR@0.1%FPR 0.1172. Its candidate-side finite-tail readout reaches 60/512 members before the first false positive in this packet. Table VI records the missing surfaces for the high readout: the packet lacks a portability surface and an action-compatible consumer boundary.

The SD/CelebA img2img cache sets the portability boundary. On the 25/25 portability-test cache, output-cloud geometry reaches AUC 0.7888 with zero strict-tail recovery ($n_0 = 25$). A smaller 10/10 stability cache is too small for portability evidence. We report output-cloud geometry as a controlled response-cloud observable in a candidate boundary case: label shuffles collapse to random-level AUC, shared-position and

TABLE V

CALIBRATION EVIDENCE PACKETS WITH BOUNDED CLAIM ROLES. EACH ROW SPECIFIES THE EVIDENCE TIER AND METRIC PROVENANCE. BOTH LOW-FPR COLUMNS ARE FINITE-TAIL PACKET READOUTS TIED TO THE NONMEMBER COUNT n_0; TPR@0.1%FPR ALSO REPORTS RECOVERED TRUE POSITIVES OVER MEMBER COUNT AND THE FALSE-POSITIVE CAP IMPLIED BY n_0. POINT ROWS REPORT SOURCE-DOCUMENTED VALUES; REPLAY-DERIVED INTERVALS ARE REPORTED IN `METRIC_UNCERTAINTY.CSV` AND ADJACENT TEXT, OUTSIDE THE COMPACT TABLE.

Role/tier	Method	AUC	ASR	TPR@1%FPR (n_0 tail)	TPR@0.1%FPR (TP/mem; FP cap)	Claim role
BB main / point	recon DDIM public-100 step30 [6]	0.837	0.740	0.220	0.110 (11/100; cap 0)	black-box proxy point estimate; no row-score replay or downstream claim
GB main / row	PIA GPU512 baseline	0.841	0.786	0.059	0.012 (6/512; cap 0)	bounded gray-box row replay with provenance caveat
GB comp. / row	PIA + stochastic dropout	0.828	0.768	0.053	0.010 (5/512; cap 0)	provisional defense comparator
WB comp. / point	GSA 1k-3shadow [7]	0.998	0.990	0.987	0.432 (432/1000; cap 1)	upper-bound white-box comparator
WB bridge / target	GSA against DPDM W-1 [7], [22]	0.489	0.499	0.009	0.000 (0/1000; cap 1)	target-score defense bridge

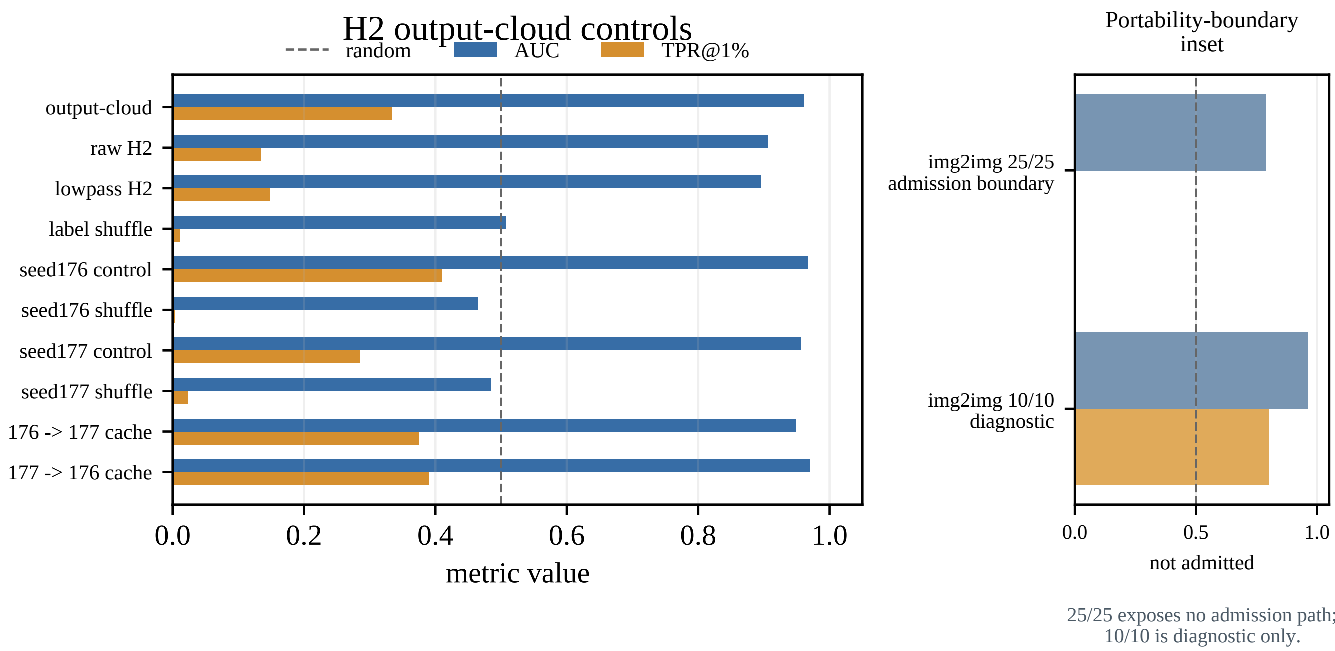

Fig. 5. Candidate-side admission stress test for H2 output-cloud geometry. The H2-family metric is above chance and stays candidate-only: it collapses under label shuffle and persists under seed-offset control, seed stability, and same-family cache-reuse checks. SD/CelebA img2img portability rows are shown as a boundary inset: the 25/25 packet leaves the admission path absent, and the 10/10 row is diagnostic only.

same-family cache checks show same-family stability, and the img2img packet does not supply portability or downstream audit admission.

IX. ACTIVATION FINGERPRINTS: STABLE AUC, FRAGILE TAIL

H2 demonstrates that high AUC alone does not satisfy portability and consumer-boundary gates. A second boundary case asks whether a mechanistically distinct white-box observable can survive diagnostic admission when its aggregate signal is clean but its low-FPR tail collapses under scale-up.

A. Motivation

The admitted white-box rows are gradient-based (GSA, DPDM bridge). GSA reaches AUC 0.998 and TPR@1%FPR 0.987, close to a saturation ceiling for the CIFAR-10 target. H2 is a response-contract observable. A natural measurement question is whether internal UNet activations—neither gradient nor loss nor response-contract—carry a membership signal that could serve as a second independent white-box family.

B. Method

We register forward hooks at four UNet sites on the E3-calibrated CUDA DDPM 800k target (UNet ch=128, ch_mult=[1,2,2,2], num_res_blocks=2): late_down (deepest

TABLE VI
H2 OUTPUT-CLOUD ADMISSION-DECISION SUMMARY.

Question	Evidence	Contract decision
Metric signal	Output-cloud AUC 0.9615; raw/lowpass H2 diagnostics AUC 0.9057/0.8957.	Candidate observable; consumer boundary and delta remain unresolved.
Same-family controls	Label-shuffle AUC 0.5076; shared-position, seed, and same-family cache-reuse AUC 0.9489–0.9705.	Same-family reuse support; portability unresolved.
Admission boundary	SD/CelebA img2img 25/25 AUC 0.7888 with zero strict-tail recovery ($n_0 = 25$).	Portability admission path absent in the current packet.

TABLE VII
H1 ACTIVATION-SUBSPACE SCOUT AND SCALE-UP RESULTS.

Stage	N	AUC	TPR@1%FPR	TPR@0.1%FPR
Scout (3 shadows)	64	0.874	0.484	0.000
Scale-up (3 shadows)	128	0.873	0.055	0.000
Label shuffle (N=128)	128	0.486	–	–
Ablation (drop late_down)	128	0.872	–	–
Ablation (drop mid_0)	128	0.873	–	–
Ablation (drop mid_1)	128	0.875	–	–
Ablation (drop early_up)	128	0.875	–	–

ResBlock before middle), mid_0, mid_1, and early_up (first ResBlock after middle). At each site and timestep $t \in \{100, 400, 700\}$, we extract per-channel statistics: mean absolute activation μ_{abs}, channel variance σ^2, and activation sparsity (fraction below $0.01 \cdot \sigma$). Per-site scalar aggregates are combined with PCA-6 projection of the stacked per-channel vectors.

Three shadow calibrations (seeds 42–44) train a balanced logistic regression on member/nonmember labels. The evaluation uses the same E3 CIFAR-10 50/50 member/nonmember split at $N = 64$ and $N = 128$.

C. Results

Table VII reports the main measurements. AUC is stable: ΔAUC $= +0.001$ from $N = 64$ to $N = 128$. Label-shuffle control returns random-level AUC (0.486), confirming the aggregate signal is not a dimensionality artifact. Per-site ablation removes each of the four hook sites in turn; all ablated AUCs remain within $[0.867, 0.879]$, ruling out a single-site artifact.

We decompose the signal by statistic type. Under the combined $\mu_{\text{abs}}+\sigma^2$ feature set, the LR reaches AUC 0.839. Using only μ_{abs} yields AUC 0.800; using only σ^2 yields AUC 0.818. Either statistic alone captures 95–97% of the combined signal—they are redundant carriers. Sparsity features alone yield AUC 0.500 (chance), consistent with negligible coefficient share ($<$1%). The signal is thus distributed across statistic types as well as across UNet sites.

D. Admission Failure: Low-FPR Collapse

Despite stable aggregate AUC, H1 fails low-FPR admission. TPR@1%FPR collapses from 0.484 ($N = 64$) to 0.055 ($N = 128$). At $N = 128$ clean nonmembers, the 1% FPR operating point requires $\sim$1.28 false positives; the highest-scoring nonmembers consistently rank above most members. TPR@0.1%FPR remains zero at both sample sizes, though $N = 128$ is below the rule-of-three threshold for calibrated 0.1% FPR claims.

E. Channel Intervention: Causal Test

Per-channel t-tests identify $\sim$4% of channels with significant member/nonmember differences ($p < 0.01$). To test whether these channels are causally responsible for the H1 score, we perform targeted channel knockout: zeroing specific channels during the forward pass and re-evaluating with both retrained and frozen scorers.

Matched-count (both delete exactly 10 channels): targeted top-10 ΔAUC=$+0.008$, within 1σ of random-10 deletion (μ_Δ=$+0.018$, σ=0.015).

Matched-percent (both delete $\sim$40 channels, 4% of total): targeted top-4% ΔAUC=$+0.008$, and random 4% matched deletions produced a similar mean effect (μ_Δ=$+0.004$, σ=0.019 across 30 seeds), with the targeted–random difference not statistically significant (Cohen's d=0.21, p=0.26, 95% CI on difference $[-0.003, +0.011]$). The 95% CI is narrow and centered near zero, indicating that the data are compatible with practical equivalence at the observed effect size. We note that 30 seeds provides limited power for small effects (80% power at d=0.2 requires $\sim$350 seeds per group); we therefore interpret this result as evidence against a large targeted–random disparity rather than as a definitive proof of exact equivalence. Neither targeted nor random 4% knockout meaningfully reduces H1 AUC, confirming that the signal is distributed and redundant rather than controlled by a compact channel subset.

Frozen scorer: random 4% knockout drops frozen-scorer AUC by 0.138, but retraining recovers the loss—the signal family is redundant.

These results rule out a sparse-channel causal interpretation at the level of individually significant channels. However, full-site knockout reveals a clear causal gradient across UNet depth. Table VIII reports the effect of zeroing all channels at each site, on both the DDPM 800k and DDIM 750k checkpoints.

The causal gradient is monotonic with UNet depth and identical across checkpoints: late_down $\gg$ mid_0 $>$ mid_1 $>$ early_up. Zeroing late_down collapses AUC to chance (0.500) on both checkpoints, while zeroing early_up produces only a mild drop. The most statistically visible channels (concentrated in early_up, 4.7% significant at p<0.01) are at the *least* causally important site. This separation—statistical visibility highest in early_up, causal importance highest in late_down—is the central mechanistic finding: membership-correlated channels are not membership-causal channels.

Table IX reports the complete 4×3 spatial–temporal knockout grid on both checkpoints. The causal locus is sharply concentrated: early denoising (t=100) at the bottleneck and deepest downsampling layers (mid_0, late_down) accounts for

TABLE VIII

Full-site knockout causal gradient. Zeroing all channels at each UNet site and re-evaluating H1. The causal importance follows a monotonic depth gradient: deepest downsampling representations are indispensable, while early upsampling representations are largely dispensable. The pattern is identical across both checkpoints.

Site	DDPM 800k		DDIM 750k	
	AUC	ΔAUC	AUC	ΔAUC
Baseline (no KO)	0.874	–	0.888	–
late_down (deepest)	**0.500**	**+0.374**	**0.500**	**+0.388**
mid_0	0.625	+0.249	0.623	+0.265
mid_1	0.766	+0.108	0.766	+0.122
early_up (first up)	0.797	+0.077	0.786	+0.102

TABLE IX

Spatial–temporal causal grid. ΔAUC when zeroing all channels at each site×timestep. Early denoising (t=100) at the bottleneck (mid_0) and deepest layers (late_down) carries essentially all membership signal. The pattern is identical across DDPM 800k and DDIM 750k.

Site	DDPM 800k ΔAUC			DDIM 750k ΔAUC		
	t=100	t=400	t=700	t=100	t=400	t=700
late_down	**+0.138**	+0.039	+0.006	**+0.240**	+0.009	+0.025
mid_0	**+0.149**	+0.002	+0.008	**+0.202**	+0.015	+0.013
mid_1	+0.087	+0.011	−0.007	+0.090	−0.004	+0.009
early_up	+0.078	+0.032	+0.007	+0.099	−0.007	+0.007

TABLE X

Fine temporal grid knockout (8 timesteps). DDPM training produces temporally distributed signal: 0 of 16 individual knockouts exceed Δ=0.05. DDIM training produces temporally concentrated signal: all 16 knockouts exceed Δ=0.09, with late_down@t=100 destroying 26% of AUC.

Checkpoint (baseline)	ΔAUC (baseline − knockout)							
	t=50	t=100	t=150	t=200	t=300	t=400	t=600	t=8
late_down								
DDPM 800k (0.833)	**+0.029**	+0.010	−0.019	+0.022	+0.008	+0.004	+0.006	−0.0
DDIM 750k (0.854)	+0.166	**+0.221**	+0.130	+0.134	+0.120	+0.131	+0.094	+0.1
mid_0								
DDPM 800k (0.833)	−0.021	+0.009	+0.002	+0.003	0.000	−0.024	+0.021	+0.0
DDIM 750k (0.854)	+0.118	+0.156	+0.138	+0.147	**+0.194**	+0.120	+0.127	+0.1

nearly all membership signal. At t=400 and t=700, zeroing any site produces negligible or null effect—the signal is already encoded by early denoising. The pattern is identical across independently trained checkpoints.

F. Fine Temporal Grid: Training Procedure Controls Signal Geometry

The spatiotemporal grid (4×3) showed sharp causal concentration at t=100. A natural question: is this a genuine concentration, or an artifact of coarse temporal sampling (3 points) that amplifies apparent per-timestep importance? The LR classifier with only 3 temporal features may overstate individual timestep contributions if the true distribution is smoother.

To test this, we ran an 8-timestep fine grid ($t \in \{50, 100, 150, 200, 300, 400, 600, 800\}$) on both checkpoints, zeroing all channels at each site×timestep individually. Table X reports ΔAUC for each knockout relative to the 8-timestep baseline.

The contrast is stark. On DDPM, with 8-timestep resolution the individual knockout effects nearly vanish: the DDPM baseline is 0.833, and the maximum Δ across all 16 (site, timestep) cells is +0.029—seven times *smaller* than the 3-timestep max of +0.149. The LR classifier compensates for individual timestep knockout by drawing on redundant temporal features.

On DDIM, the opposite occurs. Individual knockout effects remain large even at 8-timestep resolution: all 16 cells exceed Δ=0.09, and late_down@t=100 destroys Δ=0.221 (26% of baseline AUC). The DDIM signal is temporally *concentrated*, not distributed.

This reveals a property we had not anticipated: **membership signal geometry is a product of the training procedure, not a fixed property of the UNet architecture.** DDPM training (stochastic, full 1000-step forward process) produces temporally distributed, redundant signal. DDIM training (deterministic inference objective) produces temporally concentrated signal vulnerable to single-timestep knockout.

The channel-level distribution (no single channel is causal) holds for both checkpoints. We further confirmed this on DDIM 750k: mu_abs-only and var-only features each individually capture 99.8% of the combined AUC (mu_only AUC 0.893, var_only AUC 0.895, combined 0.895), and the correlation–causation dissociation replicates: early_up@t=100 has 14.1% significant channels (vs. DDPM's 4.7%) yet remains the least causally important site.

Cross-dataset validation (CIFAR-100). To test whether DAAB is specific to CIFAR-10, we ran the 4-site × 3-timestep spatiotemporal grid on a CIFAR-100 DDPM with the same UNet architecture. The baseline AUC is 0.835, confirming that activation-subspace membership signal exists across datasets. However, the spatiotemporal geometry differs. On CIFAR-10, the causal gradient is sharply hierarchical: late_down@t=100 (Δ=+0.138) ≫ mid_0@t=100 (+0.149) > mid_1 (+0.087) > early_up (+0.078), with t=400 and t=700 causally negligible. On CIFAR-100, all four sites and all three timesteps exhibit substantial knockout effects: late_down $\in [+0.137, +0.240]$, mid_0 $\in [+0.177, +0.200]$, mid_1 $\in [+0.112, +0.230]$, early_up $\in [+0.129, +0.254]$. The site gradient is flatter and the temporal peak shifts from t=100 to t=400. We further ran an 8-timestep fine temporal grid on CIFAR-100 (baseline AUC 0.811). The result is striking: CIFAR-100 DDPM produces temporally *concentrated* signal, with max $|\Delta|$=0.167 at late_down—5.7× larger than CIFAR-10 DDPM (0.029) and comparable to DDIM on CIFAR-10 (0.221). Table XI summarizes the three-way comparison.

DAAB as a phenomenon—real activation-subspace membership signal with channel-level distribution—generalizes across datasets. However, the temporal distribution of that

TABLE XI
THREE-WAY TEMPORAL DISTRIBUTION COMPARISON (8 TIMESTEPS). TEMPORAL SIGNAL CONCENTRATION DIFFERS ACROSS BOTH TRAINING PROCEDURES AND DATASETS. CIFAR-10 DDPM YIELDS DISTRIBUTED SIGNAL; DDIM TRAINING AND CIFAR-100 PRODUCE CONCENTRATED SIGNAL. THE MECHANISM IS UNDERDETERMINED WITH TWO DATASETS (N=2)—CLASS COUNT, SAMPLES-PER-CLASS, AND TRAINING-PROCEDURE INTERACTIONS ARE CANDIDATES FOR FUTURE INVESTIGATION.

Configuration	late_down max $\|\Delta\|$	mid_0 max $\|\Delta\|$	Pattern
CIFAR-10 DDPM 800k	0.029	0.024	Distributed
CIFAR-10 DDIM 750k	0.221	0.194	Concentrated
CIFAR-100 DDPM	**0.167**	**0.157**	**Concentrated**

signal is not a fixed property: it depends on *both* the training procedure (DDPM vs. DDIM) *and* the dataset (CIFAR-10 vs. CIFAR-100), with the two factors confounded in the current design (class count, samples-per-class, and training duration all covary). We report the empirical observation without committing to a single causal attribution; future work with additional datasets and controlled interventions is needed to isolate the mechanism.

G. Diagnostic Lesson: Distributed Activation-Amplitude Bias

H1 reveals a **Distributed Activation-Amplitude Bias (DAAB)**: a replicated, activation-magnitude-dominated membership trace whose aggregate signal is distributed across channels and statistic types, yet whose causal core concentrates in the deepest downsampling representations (late_down) rather than in the sites where individual channels appear most statistically significant (early_up). The signal is redundant at the per-channel level—targeted knockout of membership-correlated channels does not selectively suppress H1—but is not fully distributed: the deepest layers are causally indispensable.

H1 separates four levels most MIA evaluations conflate: (1) existence—real (AUC 0.84–0.88); (2) characterization—activation magnitude, distributed; (3) causal localization—not possible (targeted KO $\leq$ random KO); (4) forensic admission—not satisfied (TPR@1% collapses). The fine temporal grid (Table X) adds a fifth finding: (5) training procedure controls temporal signal geometry—DDPM yields temporally distributed signal, DDIM yields temporally concentrated signal. Cross-dataset validation on CIFAR-100 (AUC 0.835) adds a sixth: (6) dataset complexity controls spatiotemporal geometry—CIFAR-100 shows a flatter site gradient and broader temporal distribution than CIFAR-10, confirming that DAAB generalizes as a phenomenon while its spatial and temporal distribution varies with both training procedure and dataset.

The core insight is: *"Real signal does not imply causal localization; causal non-localization does not imply forensic admission."*

A second insight emerges from the DDPM–DDIM dissociation: *"Membership signal geometry is a product of the training procedure, not a fixed property of the model architecture."*

H1 therefore enters the claim register as **candidate-positive / low-FPR-fragile (DAAB)**, not as admitted evidence.

TABLE XII
ALLOWED AND BLOCKED CLAIMS FOR KEY BOUNDARY CASES.

Case	Allowed claim	Blocked claim
GSA	White-box gradient upper-bound comparator (AUC 0.998) under scope-bounded admission.	Universal MIA success; black-box or gray-box audit evidence.
H1/DAAB	Internal activations carry replicated, mechanistically characterized membership signal (AUC 0.84–0.88); channels most correlated with membership are not causal bottlenecks.	Admitted low-FPR forensic evidence; compact editable leakage target; any TPR@0.1%FPR claim.
CLiD	Prompt-conditioned AUC collapses from 1.000 to 0.586 under prompt-neutral control; diagnostic example of spurious signal.	Reliable black-box MIA; any bootstrap CI or p-value on ΔAUC.
H2 cloud	Within-family output-cloud AUC 0.962; candidate observable with unresolved portability.	Admitted black-box evidence; portable audit claim.
scnet	$54\times$ capacity increase produces no meaningful MIA gain (ΔAUC=0.003).	Proof that capacity never increases leakage; extrapolation to production models.
MoFIT	Strong external black-box signal exists (AUC 0.942); public score-file boundary case.	Internally admitted evidence; second independent response asset.

We do not release H4 (post-training subspace editing defense): channel intervention experiments show no compact causal edit target exists. Targeted knockout of membership-correlated channels does not suppress H1 more than random deletion. The activation signal is not locally editable by sparse channel removal.

X. NEGATIVE AND SUPPORT EVIDENCE

Bounded non-admitted routes are retained when they answer a route-selection question. Table XIII records each route's packet or provenance stop, first blocker, and blocked claim, keeping score-bearing rows separate from metadata-only stops. The routes cover ReDiffuse [19], the CommonCanvas response surface associated with CopyMark materials [23], MIDST [24], and Tracing the Roots [18], plus MoFit, inspected through public score-like text files as a support boundary check [20]. For STL-10, a generated route sidecar separates the strict 50k/50k DDIM reproduction rows from the fixed-subset overfit setting. The strict rows remain bounded-negative. The overfit rows are support-only memorization checks.

A. Public Score-File Boundary: MoFit

MoFit uses public COCO text files for a paper-side score-file recheck. The files lack target identity, explicit row IDs, official metrics, consumer use, and a surface-delta control. Using public file order as a diagnostic index over 500 train-position and 500 test-position entries, the best check ($\alpha = 0.55$) reports AUC 0.941948, ASR 0.883, TPR@1%FPR 0.488, and TPR@0.1%FPR 0.324; permutation labels are random-level (mean AUC 0.500805, 95% interval [0.465629, 0.535924]). The readouts are diagnostics over public file positions. All

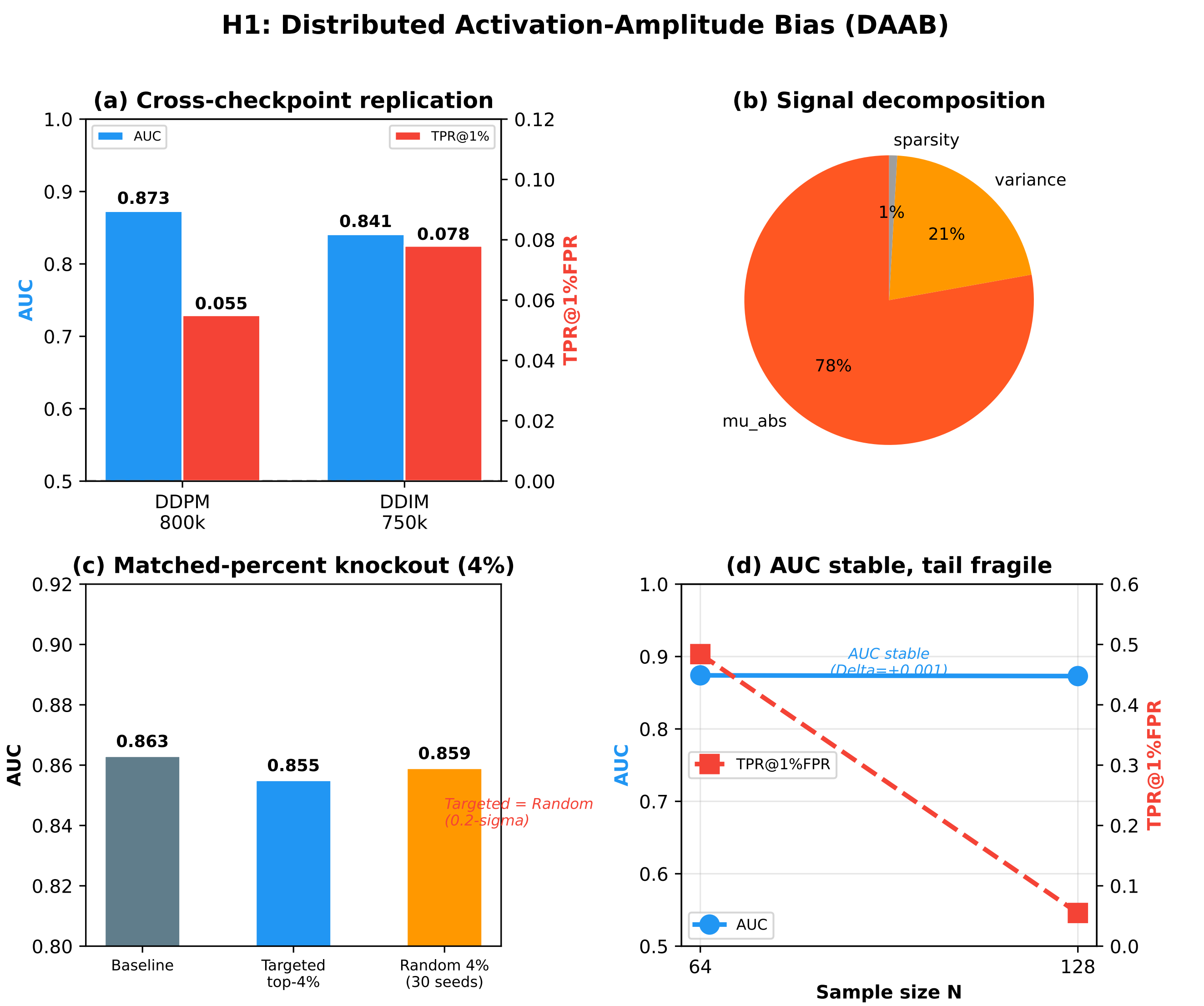

Fig. 6. **H1 Distributed Activation-Amplitude Bias.** (a) Cross-checkpoint replication: AUC remains 0.84–0.88 across DDPM 800k and DDIM 750k. (b) Signal decomposition: activation magnitude (μ_{abs}) carries 78% of LR coefficient mass. (c) Matched-percent knockout: neither targeted top-4% nor random 4% deletion meaningfully reduces AUC, confirming distributed redundancy. (d) AUC stays stable under scale-up while TPR@1%FPR collapses, demonstrating the gap between aggregate separability and forensic admissibility.

six gates are non-Pass, so MoFit is a support-only score-file diagnostic.

The table gives each route a decision role. Bounded negatives close tested routes; feature, source-confounded, and metadata rows supply boundary checks.

XI. Discussion and Limitations

A diffusion MIA score needs a packet with fixed target identity, split semantics, row coverage, metric provenance, finite-tail denominator, and consumer boundary before it can support audit wording. Missing surfaces leave high AUC open to source-label shortcuts, prompt-conditioned auxiliary features, non-replayable scores, or finite-tail artifacts.

To make the protocol directly actionable, Table XIV provides a concrete checklist: for any proposed MIA claim, the evaluator checks each gate in fixed order and records the first missing surface.

H2 is non-admitted because it lacks an independent response asset and a consumer-boundary path. MoFit is non-admitted because target identity, explicit row IDs, certified row binding, consumer boundary, and surface delta are unresolved. The current MoFit packet supplies stress coverage.

Selected-corpus claims require a larger frozen distinct-surface corpus. Coding-reliability claims require independent multi-reviewer labels. The current set of 24 boundary cases is selected to test claim-state behavior across access levels and failure modes; it is not a systematic or exhaustive benchmark, and frequency estimation requires separate corpus sampling. We do not claim that the observed proportions generalize to the broader MIA literature.

TABLE XIII

NEGATIVE/SUPPORT PACKETS ARE ROUTE-LOCAL CLAIM DECISIONS.

Route	Packet/readout	First blocker	Unsupported stronger wording
ReDiffuse STL-10	Strict 300K/500K rows stay random-level (AUC 0.5025–0.5140); fixed-subset overfit reaches AUC 0.9987–1.0.	Strict route is negative; overfit changes target, split, and training surface.	Unsupported: strict reproduction success or general method refutation.
CommonCanvas	50/50 response packet; denoising-loss AUC 0.5148.	Route-local negative; no portable second-contract evidence.	Unsupported: portable second asset.
MIDST	Best available score AUC 0.5981.	Tabular benchmark surface and weak image-diffusion relevance.	Unsupported: image-diffusion audit evidence.
Tracing the Roots	Feature packet AUC 0.8158.	Feature tensor only.	Unsupported: raw image audit row.
MoFit COCO	Support-only score-file check: all six gates remain non-Pass; AUC 0.9419 and TPR@1%FPR 0.488.	Numeric rows lack explicit row IDs and immutable public target binding.	Unsupported: second public asset or admitted row.
Separately supplied SD ReDiffuse	Replay AUC 0.7103; source-only AUC 1.0.	Source-label separation.	Unsupported: same-distribution proof.
Fixed-search trace	17 metadata observations; no evidence+metric pass.	Metadata-only protocol.	Unsupported: literature-wide rate.
Source-query controls	10 source-query observations, separate from corpus counts.	Row-set control only.	Unsupported: field-coverage rate.

TABLE XIV

CLAIM-ADMISSION CHECKLIST. FOR A PROPOSED MIA CLAIM, EVALUATE EACH GATE IN FIXED ORDER. THE FIRST NON-PASS GATE IS THE FIRST BLOCKER; THE CLAIM NARROWS TO THE STATE SUPPORTED BY PRECEDING PASS GATES.

Order	Gate	Check
G1	Target identity	Fixed architecture, checkpoint hash, and training provenance?
G2	Split semantics	Immutable member/nonmember row manifests (not inferred from source pools)?
G3	Score/response	Per-sample scores, responses, or features for all labeled rows?
G4	Metric provenance	AUC/TPR replayable from row-level data, or source-documented with provenance?
G5	Consumer boundary	Can the named consumer access all required artifacts?
G6	Surface delta	Does a claimed second asset change model, data, modality, or response contract?

The H1 spatiotemporal causal findings are demonstrated on a single architecture family (CIFAR-10 U-Net DDPMs at 750k–800k training steps) with two independently trained checkpoints. Whether the causal locus (mid_0 + late_down at t=100) generalizes to other architectures (DiT, transformer-based diffusion), datasets (higher-resolution, multi-modal), or training regimes (larger models, longer training) remains an open question. The 30-seed knockout analysis has limited power for small effects; we report the 95% CI and interpret the result as evidence against a large targeted–random disparity rather than as proof of exact equivalence. Point-role claims require row or target-score arrays.

A. Release and Validity Threats

Reusable diffusion MIA artifacts should expose target/checkpoint identities, immutable splits, row-level scores or responses, metric code or JSON, finite-tail denominators, hashes or commit IDs, and intended consumer language. The reportable bundle is bounded by permissions and available assets: rows come from local score/response replay or source-documented point estimates. The audit-replay claim stays on the exposed surface. Low-FPR tails remain finite empirical packet readouts tied to nonmember denominators, and the fault-injection cases exercise the release renderer and phrase guard.

The protocol measures exposed-surface sufficiency. Ground-truth difficulty stays separate: a packet could pass all six gates and still mislead if the member/nonmember split is trivially separable, the observable was chosen after inspection, or a known confound is omitted. When a packet is shared, include the claim trace as an immutable snapshot: a reviewer maps each headline claim to its trace row and checks the source-provenance hashes, generator command, commit identity, and availability tier. In the current packet, the provenance table is a local packet-identity snapshot; dirty tree state and local paths identify the review snapshot, and the public-release provenance gate remains unresolved.

Gate labels are frozen protocol labels over selected decisions. Inter-rater measurements and prevalence samples require independent recoding. C14 is a selected thirteen-row stress packet; field-level measurement requires a preregistered larger corpus and independent labels.

XII. CONCLUSION

The strongest lesson from these boundary cases is that diffusion MIA evidence has states. H1 reveals Distributed Activation-Amplitude Bias: internal U-Net activations carry a replicated, mechanistically characterized membership trace that is channel-level distributed and universal across training procedures, yet whose temporal distribution depends on the training objective—DDPM stochastic training produces temporally distributed signal (max individual ΔAUC=0.029), while DDIM deterministic training produces temporally concentrated signal (ΔAUC=0.221, a 7.6$\times$ difference). The signal is non-localizable under channel knockout and too fragile at low FPR for forensic admission. CLiD shows that perfect separation can be spurious under a prompt-control change. H2 shows that high response-surface AUC can fail portability. MoFIT shows that strong external scores exist but require row binding and target identity before supporting a bounded audit claim. The admitted rows—GSA, PIA, Recon—confirm that genuine membership signal exists across access levels when the evidence contract is satisfied. Taken together, these cases argue that membership inference is not only a score-ranking problem: a diffusion MIA score becomes evidence only after its claim boundary is admitted.

REFERENCES

[1] N. Carlini, J. Hayes, M. Nasr, M. Jagielski, V. Sehwag, F. Tramèr, B. Balle, D. Ippolito, and E. Wallace, "Extracting training data from diffusion models," in *Proceedings of the 32nd USENIX Security Symposium*, 2023, pp. 5253–5270. [Online]. Available: https://www.usenix.org/conference/usenixsecurity23/presentation/carlini

[2] G. Somepalli, V. Singla, M. Goldblum, J. Geiping, and T. Goldstein, "Diffusion art or digital forgery? investigating data replication in diffusion models," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 6048–6058.

[3] J. Duan, F. Kong, S. Wang, X. Shi, and K. Xu, "Are diffusion models vulnerable to membership inference attacks?" in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 8717–8730. [Online]. Available: https://proceedings.mlr.press/v202/duan23b.html

[4] F. Kong, J. Duan, R. Ma, H. T. Shen, X. Shi, X. Zhu, and K. Xu, "An efficient membership inference attack for the diffusion model by proximal initialization," in *The Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: https://openreview.net/forum?id=rpH9FcCEV6

[5] S. Zhai, H. Chen, Y. Dong, J. Li, Q. Shen, Y. Gao, H. Su, and Y. Liu, "Membership inference on text-to-image diffusion models via conditional likelihood discrepancy," in *Advances in Neural Information Processing Systems*, 2024. [Online]. Available: https://openreview.net/forum?id=DztaBt4wP5

[6] Y. Pang and T. Wang, "Black-box membership inference attacks against fine-tuned diffusion models," in *Network and Distributed System Security Symposium*, 2025.

[7] Y. Pang, T. Wang, X. Kang, M. Huai, and Y. Zhang, "White-box membership inference attacks against diffusion models," *Proceedings on Privacy Enhancing Technologies*, vol. 2025, no. 2, pp. 398–415, 2025. [Online]. Available: https://petsymposium.org/popets/2025/popets-2025-0068.php

[8] C. Herley and P. C. van Oorschot, "SoK: Science, security, and the elusive goal of security as a scientific pursuit," in *2017 IEEE Symposium on Security and Privacy*, 2017, pp. 99–120.

[9] E. Van Der Kouwe, G. Heiser, D. Andriesse, H. Bos, and C. Giuffrida, "Benchmarking flaws undermine security research," *IEEE Security & Privacy*, vol. 18, no. 3, pp. 48–57, 2020.

[10] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivière, A. Beygelzimer, F. d'Alché Buc, E. Fox, and H. Larochelle, "Improving reproducibility in machine learning research (a report from the neurips 2019 reproducibility program)," *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021. [Online]. Available: https://jmlr.org/papers/v22/20-303.html

[11] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership inference attacks against machine learning models," in *2017 IEEE Symposium on Security and Privacy*, 2017, pp. 3–18.

[12] S. Yeom, I. Giacomelli, M. Fredrikson, and S. Jha, "Privacy risk in machine learning: Analyzing the connection to overfitting," in *2018 IEEE 31st Computer Security Foundations Symposium*, 2018, pp. 268–282.

[13] N. Carlini, S. Chien, M. Nasr, S. Song, A. Terzis, and F. Tramèr, "Membership inference attacks from first principles," in *2022 IEEE Symposium on Security and Privacy*, 2022, pp. 1897–1914.

[14] J. Zhang, D. Das, G. Kamath, and F. Tramèr, "Position: Membership inference attacks cannot prove that a model was trained on your data," in *2025 IEEE Conference on Secure and Trustworthy Machine Learning*, 2025, pp. 333–345.

[15] D. Das, J. Zhang, and F. Tramèr, "Blind baselines beat membership inference attacks for foundation models," 2024, arXiv:2406.16201; accepted to DATA-FM at ICLR 2025 and IEEE DLSP Workshop 2025. [Online]. Available: https://arxiv.org/abs/2406.16201

[16] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 6840–6851. [Online]. Available: https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html

[17] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 10 684–10 695. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2022/html/Rombach_High-Resolution_Image_Synthesis_With_Latent_Diffusion_Models_CVPR_2022_paper.html

[18] A. Floros, S.-M. Moosavi-Dezfooli, and P. L. Dragotti, "Tracing the roots: Leveraging temporal dynamics in diffusion trajectories for origin attribution," 2024, arXiv:2411.07449v3; accepted to NeurIPS 2025. [Online]. Available: https://arxiv.org/abs/2411.07449

[19] J. Li, J. Dong, T. He, and J. Zhang, "Towards black-box membership inference attack for diffusion models," in *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267. PMLR, 2025, pp. 34 271–34 287. [Online]. Available: https://proceedings.mlr.press/v267/li25k.html

[20] J. Jeon, W. J. Kim, S. Ha, S. Son, and S.-E. Yoon, "No caption, no problem: Caption-free membership inference via model-fitted embeddings," in *The Fourteenth International Conference on Learning Representations*, 2026, arXiv:2602.22689. [Online]. Available: https://openreview.net/forum?id=KUXLrSXYPv

[21] P. Lian, Y. Cai, S. Li, and B. Bao, "Noise as a probe: Membership inference attacks on diffusion models leveraging initial noise," 2026, arXiv:2601.21628. [Online]. Available: https://arxiv.org/abs/2601.21628

[22] T. Dockhorn, T. Cao, A. Vahdat, and K. Kreis, "Differentially private diffusion models," *Transactions on Machine Learning Research*, 2023. [Online]. Available: https://openreview.net/forum?id=ZPpQk7FJXF

[23] C. Liang and J. You, "Real-world benchmarks make membership inference attacks fail on diffusion models," 2024, arXiv:2410.03640. [Online]. Available: https://arxiv.org/abs/2410.03640

[24] M. Shafieinejad, X. He, M. Alinoori, J. Jewell, S. Ayromlou, W. Pang, V. Chatrath, G. Sharma, and D. Pandya, "MIDST challenge at SaTML 2025: Membership inference over diffusion-models-based synthetic tabular data," 2026, arXiv:2603.19185. [Online]. Available: https://arxiv.org/abs/2603.19185

[25] J. Dubiński, A. Kowalczuk, F. Boenisch, and A. Dziedzic, "CDI: Copyrighted data identification in diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 18 674–18 684. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2025/html/Dubinski_CDI_Copyrighted_Data_Identification_in_Diffusion_Models_CVPR_2025_paper.html

[26] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. Daumé III, and K. Crawford, "Datasheets for datasets," *Communications of the ACM*, vol. 64, no. 12, pp. 86–92, 2021.

[27] M. Mitchell, S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, and T. Gebru, "Model cards for model reporting," in *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 2019, pp. 220–229.